

WESTERN CAROLINA, NC, USA

WMO: 722177

Lat: 35.195N

Lon: 83.865W

Elev: 1696

StdP: 13.82

Time Zone: -5.00 (NAE)

Period: 99-19

WBAN: 63811

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.8	19.1	2.6	6.7	28.1	7.5	8.6	26.9	18.5	45.2	16.4	43.3	2.1	80	0.374

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.4	90.3	73.5	88.3	72.9	86.4	72.2	76.9	85.5	75.6	84.3	74.6	82.8	4.1	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.0	134.9	80.6	72.9	130.2	79.7	72.2	127.0	79.0	41.7	85.6	40.3	84.8	39.3	82.5	84.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.1	12.2	10.5	DB	7.6	92.5	5.5	3.2	3.7	94.8	0.4	96.7	-2.6	98.5	-6.6	100.9	(4)
(5)			WB	6.8	78.8	5.5	2.0	2.8	80.3	-0.5	81.5	-3.6	82.6	-7.6	84.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	57.9	38.8	42.1	49.4	57.8	66.2	72.5	74.8	74.1	69.3	58.4	48.3	42.5		
	DBStd	14.70	10.02	9.09	8.52	7.33	6.24	3.98	3.57	3.60	4.94	7.84	7.95	8.96		
	HDD50	1164	370	249	116	18	1	0	0	0	0	19	123	268		
	HDD65	3690	812	642	485	230	64	3	0	0	21	231	504	698		
	CDD50	4058	23	27	98	253	502	675	768	747	580	279	71	35		
	CDD65	1107	0	0	1	15	99	228	304	282	151	25	2	0		
	CDH74	9725	0	1	51	341	1056	1984	2395	2290	1294	302	11	0		
	CDH80	3139	0	0	4	47	285	696	863	812	382	50	0	0		
(14) Wind	WSAvg	2.5	3.0	3.4	3.6	3.8	2.8	2.1	1.5	1.4	1.4	1.9	2.4	2.5		
(15) Precipitation	PrecAvg	65.2	6.7	6.5	6.6	5.1	5.0	5.8	5.8	4.8	4.3	3.4	4.8	6.9		
	PrecMax	82.6	13.4	11.3	11.1	9.7	11.7	11.5	9.3	9.2	8.2	7.7	8.4	10.9		
	PrecMin	43.9	3.6	3.0	3.3	2.3	1.6	2.2	2.6	1.5	0.0	0.2	1.9	3.0		
	PrecStd	9.4	2.5	2.0	2.3	1.9	2.1	2.3	1.8	2.0	2.2	2.1	1.9	2.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.0	71.8	79.4	84.1	89.9	92.7	92.7	91.4	89.8	84.7	75.2	69.3		
		MCWB	58.4	59.6	59.2	61.2	70.3	72.6	74.8	74.4	71.3	69.1	59.9	60.7		
	2%	DB	63.4	66.5	73.4	81.2	86.1	89.7	90.1	89.9	87.0	81.0	71.4	64.3		
		MCWB	56.0	56.9	58.7	61.1	67.8	72.1	74.8	74.0	70.1	65.3	57.0	58.4		
	5%	DB	59.7	63.3	69.9	78.7	83.1	87.2	87.9	87.8	84.2	77.4	67.5	61.5		
		MCWB	54.1	53.4	56.6	61.2	65.9	71.5	74.7	74.0	69.7	63.1	57.5	56.2		
	10%	DB	55.3	58.8	65.5	74.6	80.7	84.0	85.3	84.5	81.8	73.3	63.5	57.5		
		MCWB	51.3	50.9	54.7	59.8	65.6	70.5	73.8	72.9	68.3	61.9	55.4	53.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.6	62.9	64.2	68.3	74.6	76.4	79.9	78.3	74.5	72.2	66.5	63.7		
		MCDB	64.3	68.4	72.5	77.0	84.9	87.4	87.6	86.3	82.6	80.9	70.8	66.5		
	2%	WB	58.0	59.9	61.6	66.1	71.2	74.9	77.3	76.5	73.2	69.1	62.7	60.4		
		MCDB	61.8	65.3	69.3	74.9	79.3	84.7	85.6	85.1	81.8	74.8	66.6	63.3		
	5%	WB	55.4	56.4	59.3	64.2	69.6	73.6	76.0	75.4	71.9	67.0	60.1	57.5		
		MCDB	58.7	60.6	66.6	72.0	78.4	82.8	84.3	84.2	79.9	73.2	64.3	60.5		
	10%	WB	52.5	52.7	56.9	62.3	68.1	72.2	74.8	74.3	70.5	64.6	57.1	54.6		
		MCDB	55.0	56.6	63.2	70.1	76.8	80.8	82.7	82.3	77.9	70.6	61.5	56.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.2	22.5	23.9	26.7	24.8	22.7	20.4	21.7	23.7	26.9	26.1	21.1		
		MCDBR	24.1	27.5	30.6	32.7	29.8	26.4	23.8	24.9	27.8	30.5	31.0	24.1		
		MCWBR	17.0	16.7	16.3	15.6	13.4	11.5	11.3	11.7	12.6	15.8	18.3	17.0		
	5% WB	MCDBR	19.0	21.2	24.1	24.3	24.1	22.7	21.2	21.7	21.6	22.7	20.8	19.0		
		MCWBR	14.9	15.7	15.2	14.0	12.2	11.0	11.0	11.1	10.7	12.8	15.4	15.3		
(40) Clear-Sky Solar Irradiance	taub		0.304	0.312	0.344	0.385	0.439	0.468	0.505	0.511	0.405	0.363	0.326	0.303		
	taud		2.528	2.499	2.436	2.321	2.186	2.136	2.061	2.024	2.303	2.421	2.485	2.552		
	Ebn at Noon		287	297	294	285	270	261	250	246	270	274	275	281		
	Edh at Noon		26	30	34	40	47	49	53	54	39	31	27	24		
(44) All-Sky Solar Radiation	RadAvg		782	960	1320	1666	1841	1937	1840	1716	1482	1216	894	672		
	RadStd		56	116	107	114	129	139	129	90	133	138	95	66		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (34 neighbors)	+0.90	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+263	+173

Nomenclature: See separate page